CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2013 series

5054 PHYSICS

5054/42 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 (a) (i) measuring force just before it jumps
reading meter and pulling magnet at same time
force varies/not constant B1 [1]

(ii) sensible suggestion, e.g.
use of two people explained
pull slowly
repeat
video newton meter B1 [1]

(b) 5.5 ± 0.1 N unit required B1 [1]

(c) (i) axes: correct way round, labelled quantity and unit (on *y*-axis only) B1

scales: linear, not awkward
x-axis: e.g. 2 cm ≡ 1 *y*-axis: e.g. 2 cm ≡ 1 N B1

points plotted accurately within ½ small square
neat crosses or small points (in circle) B1

smooth curve of best fit drawn B1 [4]

(ii) increasing *n* decreases *F*
inverse relationship B1 [1]

(d) newton meter not sensitive enough
scale too big
no change/same reading
reading/force is too small (for this meter)/no force B1 [1]

(e) (i) new paper/second expt (thicker) as force smaller (or reverse argument)
paper that gives 3.0 N force B1 [1]

(ii) more sensitive
more readings
larger values for *F* B1 [1]

(f) **yes +** aluminium non-magnetic B1 [1]

2 (a) diagram showing paper and plain mirror
plus incident and reflected rays **OR** four roughly correct pins B1

2 pins placed on incident ray B1

pins or image (of pins) viewed in/through mirror B1

lines drawn **and** angles *i* and *r* **measured** to normal B1 [4]

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(b) sensible suggestion, e.g.
view bottom of pins
pins vertical
pins far apart, e.g. greater than 5 cm
repeat for different angles/repeat experiment
sharp pencil B1 [1]

3 **(a)** **(i)** 0.9 V cao (unit required) B1 [1]

(ii) crocodile clips
tight connections explained, e.g. wrap wire and tape B1 [1]

(iii) same value/0.9 V and needle to right B1 [1]

(b) sensible suggestion, e.g.
e.m.f./voltage too small
run down quickly/small amount of energy
voltage not steady
current too small
resistance too large B1 [1]

(c) **(i)** **1.** 2.7 (V) ecf 3 × **(a)(i)** B1 [1]
2. correct wiring in series **and** connected to voltmeter B1 [1]

(ii) **1.** 0.9 (V) ecf = **(a)(i)** B1 [1]
2. correct wiring in parallel **and** connected to voltmeter B1 [1]

4 **(a)** measures all ten together and divides by ten B1

how stops marbles moving, e.g.
in a groove
between two rulers
5 or more in a line shown touching each other B1

how ends are marked, e.g.
use of blocks
correct use of set squares B1 [3]

alternative methods:
methods of measuring one marble can score **max. 2**

measuring all 10 and averaging (B1)

technique, e.g.
set squares/blocks with one marble
circumference from:
string/paper rolled round marble then ÷ π
ink dot on marble and roll then ÷ π (B1)

(b) **(i)** 16.8(0) mm / 1.68(0) cm cao (unit required) B1 [1]

(ii) diameter (of same marble) measured more than once in different direction(s) B1 [1]

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